

TELEMETRY-DRIVEN ERGONOMIC ENERGY SYSTEMS FOR REAL-TIME MONITORING AND OPTIMIZATION

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ABSTRACT

Amusement parks rely heavily on energy to power attractions, shows, and guest experiences. This project presents a telemetry-driven interactive system that captures mechanical energy from human activity and converts it into stored electrical power for user-triggered outputs. Generators integrated with sensors collect data on rotational speed, voltage, and current to estimate power generation and system efficiency. An embedded microcontroller processes this sensor data, enabling real-time monitoring and responsiveness. The data is displayed through an interactive interface that provides real-time feedback on user energy contribution. Designed as a scalable proof of concept, this system demonstrates how telemetry and sensor-based data can enhance human-powered energy collection and support energy optimization strategies.

KEYWORDS

Human-powered energy, Interaction, Sensors, Microcontroller, Interface, Adafruit INA260

INTRODUCTION

Sustainability has become a key design objective within the themed entertainment industry, driving the adoption of technologies that reduce resource consumption and environmental impact. As the scale of interactive attractions continues to grow, so does their cumulative operational energy demand, creating a need for systems that remain engaging while minimizing dependence on external power sources and improving energy efficiency.

This work presents the design and implementation of a self-sustaining interactive attraction that harvests mechanical energy directly from human participation. The prototype system employs a manually actuated hand-crank mechanism coupled with a gear train to drive an electric generator, producing sufficient electric energy to operate an Arduino-based microcontroller unit. An integrated Adafruit INA260 continuously measures current, voltage, and power output, which enables real-time characterization of the generated electrical energy. An additional integrated AM312 PIR Motion Sensor Module measures the revolutions per minute (RPM) of the hand

crank. The acquired telemetry is processed locally and transmitted to a companion application, where guests can monitor live system performance through graphical visualizations. When a predefined power threshold is reached, an LED indicator is illuminated as the primary physical output, which further enhances the system's interactive response.

LITERATURE REVIEW

The theme park entertainment industry utilizes a large amount of energy per day to function. It is estimated that Walt Disney World in Orlando, Florida, uses more than one billion kilowatts of electricity a year to power all of its rides alone [1], with the number surely rising as the parks receive more attractions. Some theme parks around the world have already taken the initiative to be more energy efficient, such as Legoland New York with its kinetic energy dance floor used in Minifigure Skyflyer [2], or the Netherlands' theme park, Toverland, which has implemented a device to distribute energy reliably and efficiently [3].

Energy-efficient products can also be used in more accessible forms of entertainment, such as heavily utilized guest interactive experiences. Theme parks have provided an influx of guest-interactive experiences, incorporating interactive play environments and games (similar to the Donkey Kong drums at Universal's Epic Universe and the spinning barrels at Magic Kingdom). Our prototype takes advantage of these interactive environments by creating a system that can be implemented in theme parks. With Universal Studios Orlando entertaining over 21 million guests a year [4], the use of an interactive system to generate energy can be beneficial to help with the high costs associated with energy use in the United States.

MECHANICAL DESIGN

Housing Unit

The housing unit was designed using Autodesk Fusion 360 to securely house the internal gear train assembly and electronic components. The enclosure was fabricated from 7 mm laser-cut wood to achieve a rigid and manufacturable structure. Additionally, an opening was cut into the front side of the housing component to allow the hand crank to pass through to connect to the gear train assembly. The resultive design provides an efficient and cost-effective housing structure for the overall internal system.

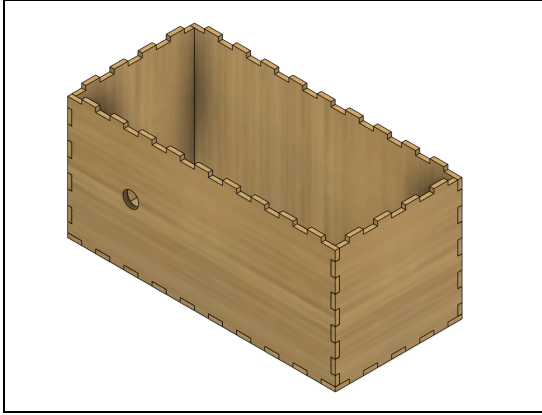


Figure 1: *CAD model of the housing unit*

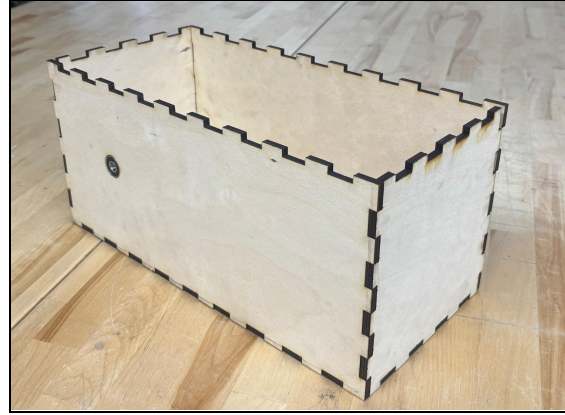


Figure 2: Physical, laser-cut housing unit

Gear Train

The relatively low rotational speed produced by manual cranking necessitated the implementation of a gear system to increase the rotational speed of the DC generator. We configured the system with an overall gear ratio of 1:9. This specific ratio was selected to better utilize the rotational speed generated through manual input while still maintaining a compact design. To accommodate the increased torque requirement associated with driving the stepped-up generator speed, the hand crank was designed with an extended length of 228.6 mm to reduce the needed input force to turn the crank.

We then calculated theoretical values as a baseline for comparison with our experimental results. For theoretical analysis, a nominal crank speed of 60 RPM was assumed. This number represents a practical and repeatable operating condition that corresponds to approximately one revolution per second that is realistically achievable for short-duration manual operation. Under this assumption, the generator rotates at approximately 540 RPM. Assuming that the generator's output voltage scales linearly with rotational speed, the theoretical no-load output voltage at this condition is approximately 4.8 V. Idealized estimates obtained by proportionality scaling the generator's rated specifications correspond to 70 W of electrical power and 14.6 A of current. These values are presented only as theoretical reference points based on our configuration and the manufacturer's specifications and do not account for electrical loading, transmission losses, generator efficiency, or the practical limitations of human-powered energy. Physical experimental testing was therefore conducted to evaluate the actual prototype performance under realistic operating conditions.

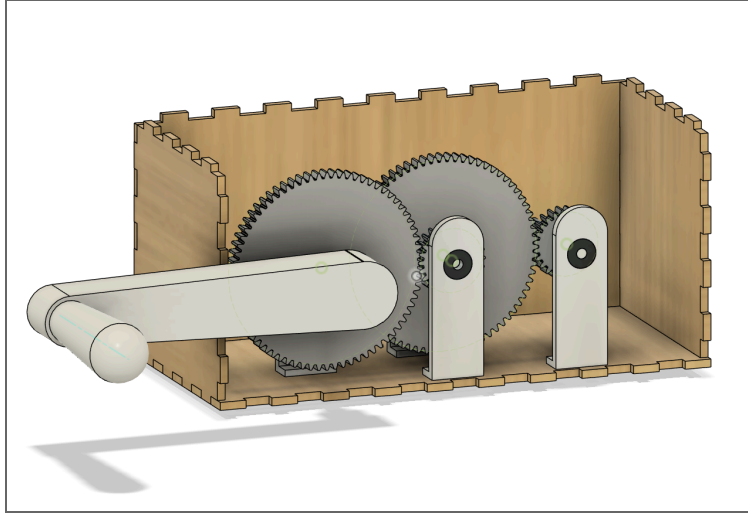


Figure 3: *CAD model of the gear train assembly inside the housing unit, shown with the front cover removed.*

As an extension of the gear train design, two iterations of the final gear train assembly were developed to evaluate different methods of mechanical/electrical energy conversion. Both designs utilized an identical output gear and overall gear ratio of 1:9 to ensure comparable performance between the two. Version 1 (V1) coupled the output gear directly to a DC motor operating as a generator. Version 2 (V2) used a custom electromagnetic generator consisting of rotating permanent magnets and a stationary coil assembly. But, after experimental testing, our data indicated that V1 consistently provided a more stable and repeatable electrical output for more reliable telemetry acquisition and system operation. Although V2 successfully generated a small, inducted voltage, the design overall produced an insufficient current at the rotational speeds achievable through manual actuation. Consequently, the generated power was not sufficient to support consistent data collection, and therefore, the DC generator configuration was adopted for our final prototype.

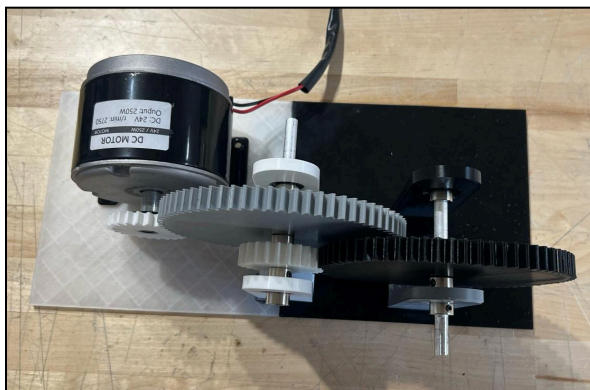


Figure 4: *Final Gear Train Assembly V1*

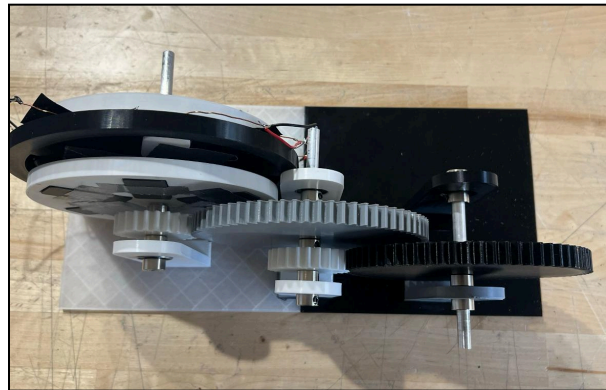


Figure 5: *Gear Train Assembly V2*

ELECTRICAL HARDWARE

Current, Voltage, and Power Sensor (Adafruit INA260):

In order to track how much energy we are collecting, we utilized a current, voltage, and power sensor (Adafruit INA260). This sensor works like a multimeter, reading the voltage across an internal 2 milliohm resistor with a 16 bit analog to digital converter (ADC). The breakout board can then accurately calculate the power generated, which is then sent to our microcontroller (Arduino UNO). The sensor calculates power by using the equation

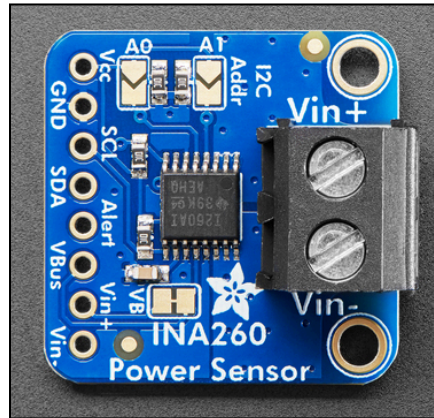


Figure 6: *Adafruit INA260 High or Low Side Voltage, Current, Power Sensor*

Microcontroller (Arduino UNO)

The microcontroller we chose to use is an Arduino UNO. This microcontroller hosts all of our code for real-time characterization of the collected energy for both the guest interactive game and for operations to monitor.

Motion sensor (Ruiwaer HC-SR501 PIR Motion Sensor):

In order to interact with our guests, we incorporated a compact passive infrared sensor (Ruiwaer HC-SR501 PIR Motion Sensor) that we used to measure how many times the handle is turned. This data is also sent to our microcontroller and will be used in the game programming.

DC Motor (AeKeatDa):

Using a 24V DC Electric Brush Motor as our main energy source, we attached a 3D-printed gear and connected it in series with our 9:1 gear ratio. The positive and negative motor ends were connected to the Adafruit INA260, with the positive end plugged into Vin+ and the negative end plugged into Vin-. For energy to be sent into the INA260 sensor we ensured that the crank was spun clockwise which forced the motor to move counter clockwise through our gear ratio. Moving the end of the motor counter clockwise allowed for the correct polarity to go throughout

the sensor. Utilizing the DC Motor allowed for a simple method of integration to connect with our power sensor.

SOFTWARE

The user interface and analysis diagrams were displayed in a React app using a client/server framework. The web application featured a homepage overview, a page for analysis including graphs with adjustable time ranges, and a page to display all the logs and measurements taken from the sensors.

An arduino was connected to a current, voltage, and power sensor (Adafruit INA260) and a motion sensor (Ruiwaer HC-SR501 PIR Motion Sensor Module) which was placed beneath the crank to measure the rotations. The rotations were stored with timestamps and precision down to the second and converted to rotations per minute (RPM). These readings were then printed to the serial monitor.

The server consisted of two main portions: the serial monitor where the arduino logged live data measurements, and the sqlite api to which the parsed data was written and could be fetched by the web application in real-time using websockets. A javascript function ran while the servers were running to parse the data as it came in and write to the sqlite database, recording timestamps as well as the data read. Readings for the motion sensor were recorded on every new HIGH reading of the sensor, while sustained HIGH readings were treated as a single event. The voltage, current, and power were read every 500ms to account for continuous and less sharp changes.

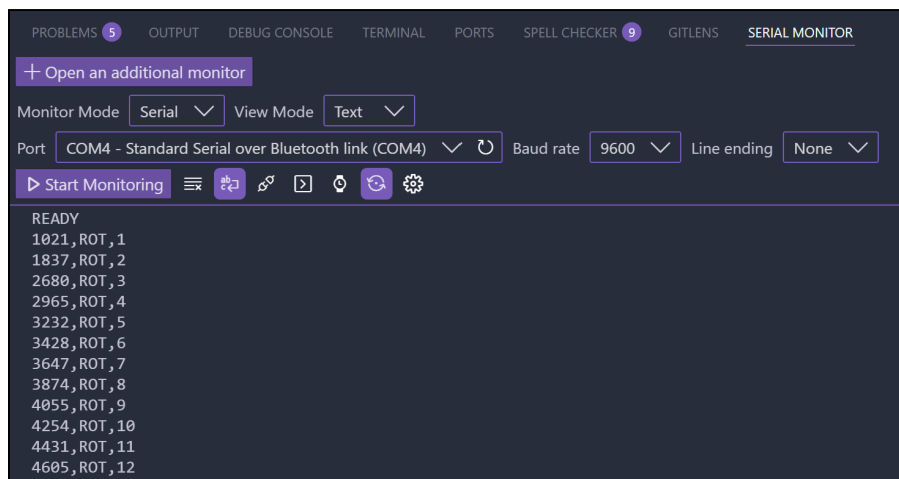


Figure 7: Live serial monitor data to be parsed by the server functions

The web application utilized websockets rather than traditional GET functions in order to allow for live data to be added without relying on refreshing the page in its entirety, allowing for users to see their immediate impact when the crank was turned to generate energy. The functions

within the react app then used the data to create graphs and interfaces which accurately reflected both the data and the trends of the data based on the last minute, day, or all time recorded events. In addition to having live data displayed in tables and graphs, the patron-facing user interface included gamified components such as pop-ups on the screen when certain records were passed or set thresholds were reached based on the data of the current day.

RESULTS

The data collected was used to help give us numerous results that were used to analyze the amount of energy generated and if it could be used in another system. As shown in figure 8 we were able to gather the amount of rotations we had for moving the hand crank as well as the power created from our model.

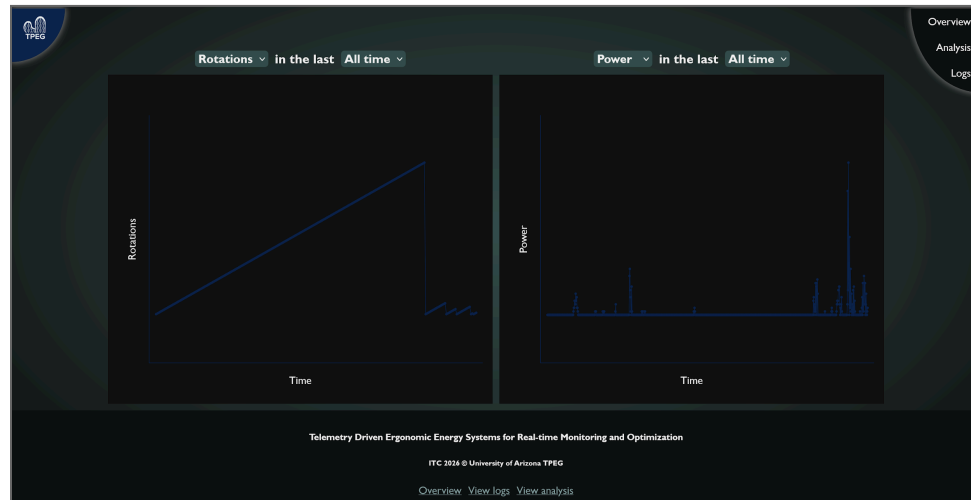


Figure 8: *Graphs showing the rotations over time and the power over time*

Both graphs feature customization options where the integral of the measured values or the instantaneous values themselves can be shown. The first graph shows the total rotations over time, where the dips seen correspond to when the system was stopped and restarted, thus resetting the totals for the run. This is an example of the demonstration of the integral of the acquired data, while the second graph shows the instantaneous values read by the sensors. The peaks correspond to the most active periods of the crank being turned while the lower values coincide with periods where the crank was turned slower, or not at all.

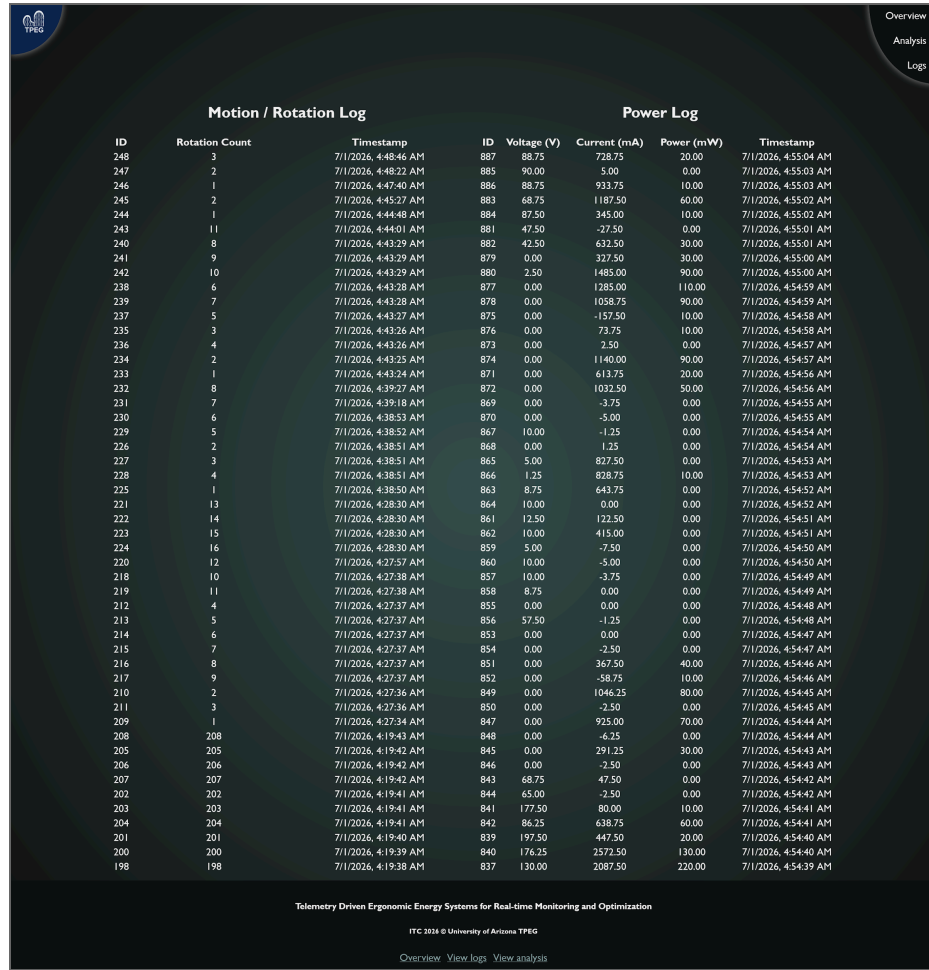


Figure 9: Website logs of rotation and power

The method tested was able to give back data going over the amount of power generated along with the current, the voltage, the rotations per minute, and number of rotations for our generator with a timestamp of when the data was taken. This data allows us to view the time guests are rotating it, when our hand crank is being utilized, and how much power we were able to generate over that time period. Even though our power was not as large as expected, future prototypes would include a large gear ratio and stronger material in our integration. The telemetry system integrated was successful in the standpoint of giving accurate data that can be used to help give insight on energy generated.

CONCLUSION

This project presents a proof-of-concept interactive system that demonstrates the potential in combining human-powered energy harvesting with embedded telemetry for themed entertainment industry applications. Through the integration of a mechanically actuated gear train, embedded sensors, and a real-time visualization interface, our prototype illustrates how

people's simple mechanical movements can be converted into measurable electrical energy while providing immediate feedback on the individual's contributions.

Although the prototype successfully demonstrated the proposed concept, the electrical power generated was limited and insufficient for large-scale energy production. Constraints on development time, budget, and the materials selected for manufacturing reduced the overall efficiency of the system, resulting in a relatively low overall power output. Despite these limitations, our project establishes a foundation for future investigation. As sustainable technologies continue to gain importance within the industry, this work demonstrates a practical starting point for integrating guest interaction with renewable energy generation and opens opportunities for broader applications in other interactive environments.

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